

Network Engineer Master Interview Guide

This guide combines scenario-based networking interview questions, Cisco CLI troubleshooting commands, and a complete HR + technical mock interview with in-depth explanations.

Section 1: Top 30 Scenario-Based Networking Interview Questions

1. A user cannot access the internet. How do you troubleshoot?

Start by checking the IP configuration using `ipconfig` or `ifconfig`. Verify the IP address, subnet mask, default gateway, and DNS. Then ping localhost, own IP, gateway, and an external IP like 8.8.8.8. If IP works but domain names fail, the issue is DNS. This structured approach isolates whether the problem is on the host, LAN, WAN, or DNS.

2. Users in one VLAN cannot reach another VLAN.

Verify that inter-VLAN routing is configured. Check VLAN assignment, trunk links, and router-on-a-stick or SVIs on the Layer 3 switch. Ensure the default gateway for each VLAN is configured and routing is enabled.

3. DHCP clients are not getting addresses.

Check whether the DHCP server is reachable, the scope has free addresses, VLANs are correctly configured, and `ip helper-address` is configured if the DHCP server is on another subnet.

4. OSPF neighbors are not forming.

Check area IDs, subnet masks, hello/dead timers, and authentication. Use `'show ip ospf neighbor'` to identify neighbor state and troubleshoot mismatches.

5. Network is slow during office hours.

Check bandwidth utilization, interface errors, duplex mismatches, and broadcast storms. Use monitoring tools to identify high-usage devices and applications.

6. Example networking scenario troubleshooting question

In an interview, explain the troubleshooting methodology: identify the issue scope, gather information, test connectivity layer by layer, check logs/configuration, isolate the fault, implement a fix, and verify resolution. This demonstrates a structured real-world troubleshooting approach.

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26. Example networking scenario troubleshooting question

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27. Example networking scenario troubleshooting question

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Section 2: Top 50 Cisco CLI Troubleshooting Commands

show ip interface brief

Displays interface IPs and status to quickly identify down interfaces.

show ip route

Displays routing table and helps verify whether a route exists.

show running-config

Shows current active configuration.

show vlan brief

Lists VLANs and port membership.

show interfaces trunk

Displays trunk ports and allowed VLANs.

show mac address-table

Shows learned MAC addresses and associated ports.

show arp

Displays IP-to-MAC mappings.

show ip ospf neighbor

Displays OSPF neighbors and their states.

show ip eigrp neighbors

Displays EIGRP neighbors.

show ip nat translations

Displays NAT translations.

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Section 3: Complete Network Engineer Mock Interview (HR + Technical)

Tell me about yourself.

Give a concise summary of your background, certifications, networking skills, lab experience, and career goals. Highlight practical exposure to routing, switching, troubleshooting, and tools.

Why do you want to work as a Network Engineer?

Show enthusiasm for networking, troubleshooting, and infrastructure reliability. Mention your interest in designing and maintaining efficient networks.

Explain the OSI model in troubleshooting.

Use the OSI model as a framework: physical layer for cables, data link for VLANs/MAC, network for IP/routing, transport for TCP/UDP ports, and application for services.

How do you troubleshoot packet loss?

Check interface errors, latency, congestion, MTU mismatch, duplex mismatch, and routing paths. Use ping, traceroute, and interface counters.

How do you handle pressure during outages?

Prioritize critical services, communicate clearly, follow troubleshooting steps methodically, escalate when necessary, and verify restoration.

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